
ABILAKSHAN VADIVEL

SURVEYOR | GIS ANALYST

Professional Interests: Land & Engineering Surveying |
Hydrographic Survey | UAV/Drone Survey | Survey Control | GIS

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Currently based in Maldives

Sri Lankan citizen

*B.Sc. (Hons) in Surveying Sciences,
specialized in GIS*

PROFESSIONAL SUMMARY

Surveyor and GIS Analyst with a B.Sc. (Hons) in Surveying Sciences, specialized in GIS, and more than three years of combined surveying and geospatial experience. Experienced in land, engineering, topographic, boundary, hydrographic, coastal, and UAV/drone surveying, including survey control, GNSS/GPS, Total Station, levelling, Ground Control Points (GCPs), field verification, and survey data processing. Skilled in AutoCAD, Civil 3D, Pix4D Mapper, ArcGIS Pro, and ArcGIS Online, with the ability to integrate field observations, drone outputs, and GIS analysis into accurate maps, plans, point clouds, elevation products, and technical deliverables.

TECHNICAL PROFILE

Surveying & GIS Skills:

Boundary & Land Surveying | Topographic & Engineering Surveying | Construction Setting-Out | Survey Control | GNSS/GPS | Traversing & Levelling | Hydrographic & Coastal Surveying | UAV Photogrammetry | GCP Establishment | CAD-to-GIS Conversion | Spatial Analysis | Land Information Systems

Software:

AutoCAD | Civil 3D | Pix4D Mapper | ArcGIS Pro | ArcGIS Online | ArcGIS Experience Builder | QGIS | PostgreSQL/PostGIS | GeoServer | CloudCompare | Google Earth Engine | MS Office

Survey Instruments & Technology:

GNSS/GPS Receivers | Total Stations | Automatic Levels | Theodolites | DJI Mavic 3 Enterprise | UAV/Drone Survey Platforms | Terrestrial Laser Scanners (TLS) | Ground Penetrating Radar (GPR)

WORK EXPERIENCE

Surveyor | Riyan Private Limited | Maldives

May 2026 - Present

- Conduct professional land, engineering, topographic, boundary/demarcation, hydrographic, coastal, and construction-related surveys for development, infrastructure, reclamation, dredging, and planning projects.
- Establish and verify survey control and Ground Control Points (GCPs), perform GNSS/Total Station/levelling observations, carry out field verification, and prepare/check survey plans and spatial records.

- Plan and execute UAV surveys using DJI Mavic 3 Enterprise with base-station setup and GCP control, and process/quality-check imagery in Pix4D Mapper to produce orthomosaics, point clouds, elevation products, maps, and technical deliverables.
- Perform GIS data preparation, spatial analysis, mapping, visualization, and web-based GIS management using ArcGIS Pro and ArcGIS Online, while coordinating technical information with engineers, planners, consultants, contractors, clients, and other project stakeholders.

**Planning Intern (Surveyor & GIS Analyst) | Riyan Private Limited
| Maldives**

Dec 2025 - Mar 2026

- Developed and edited LIS datasets in ArcGIS Pro through CAD-to-GIS conversion, land-use feature creation, attribute preparation, and spatial data quality checks.
- Published ArcGIS Online Web Maps, Web Scenes, and Experience Builder applications, and supported QGIS/PostgreSQL/PostGIS/GeoServer workflows for land-use planning and visualization.
- Supported GNSS-based boundary verification, harbour/coastal GIS tasks, and review of Pix4D orthomosaic, point-cloud, DTM, and DSM outputs.

Junior Surveyor | North Axis Surveys (Pvt) Ltd | Sri Lanka

Oct 2022 - Dec 2025

- Conducted land, boundary, topographic, and control surveys, including traversing, levelling, control establishment, land demarcation, and construction layout.
- Operated Total Stations, Sokkia Automatic Levels, and GNSS equipment; prepared survey drawings and reports using AutoCAD and GIS software.

GeoAI for Disaster Resilience - Undergraduate Fellowship | ADRIMP

Feb 2026 - May 2026

- Applied GeoAI, spatial analytics, hazard mapping, and risk-modelling concepts to disaster-resilience workflows.

SURVEYING PROJECTS

Topographic Survey & Control Establishment | North Axis (Pvt) Ltd

- Participated in a professional topographic survey for a water tank and sump project in Meesalai, Jaffna.
- Established three primary control points and collected field data for contour plan preparation.

Engineering Survey for Reservoir Construction | Academic Field Project

- Conducted engineering and topographic survey tasks for a proposed reservoir and supported access-road design using field survey data.

SELECTED GIS PROJECTS & RECOGNITION

Maldives Permanent Survey Marks (PSM) Web GIS Database

- Developing an ArcGIS Online Map Viewer and Experience Builder application to visualise Permanent Survey Mark locations and support survey-control access across Maldivian islands.

Drone-Based Coastal Analysis - Gulhi Island, South Male Atoll

- Produced a UAV-derived coastal analysis map combining drone orthomosaic, elevation/terrain information, land-use context, and spatial analysis for Gulhi Island.
- Selected among the final shortlisted map creators for the Esri India User Conference 2026 Map Gallery, with the map scheduled for showcase at the conference.

EDUCATION

B.Sc. (Hons) in Surveying Sciences, specializing in GIS | Sabaragamuwa University of Sri Lanka

- Second Class Upper Division | GPA: 3.51 / 4.00 | Faculty of Geomatics
- Faculty of Geomatics Gold Medalist for Excellent Overall Performance (2026).

PROFESSIONAL MEMBERSHIPS

- RICS Student Professional No. 0994438 | ISPRS Student Consortium | Geo Mappers

REFERENCES

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